

Lecture 11 - Implementation Aspects:

For DT systems / Filters

Introduction (System Function $\rightarrow$ DE $\rightarrow$ Imp)
 FIR
 IIR

①

We Have design the Filter $\rightarrow$ we have the system function
 $\rightarrow$ we can achieve Difference Equation From it $\rightarrow$ Implementation

For example:

$$H(z) = \frac{1+z^{-1}}{1+z^{-2}-2z^{-1}} \rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{1+z^{-1}}{1+z^{-2}-2z^{-1}}$$

$$(1+z^{-2}-2z^{-1})Y(z) = (1+z^{-1})X(z)$$

$$Y(z) + z^{-2}Y(z) - 2z^{-1}Y(z) = X(z) + z^{-1}X(z)$$

$$\xrightarrow{z^{-1}\{ \}} y[n] + y[n-2] - 2y[n-1] = x[n] + x[n-1]$$

from both sides

$$(x[n-a] \leftrightarrow z^{-a}X(z)) \quad \xRightarrow{DE} y[n] = x[n] + x[n-1] - y[n-2] + 2y[n-1]$$

we have z with positive powers

$$IF \rightarrow H(z) = \frac{z(z+1)}{z^2+1-2z}$$

$$\frac{xz^{-2}}{xz^{-2}} = \frac{1+z^{-1}}{1+z^{-2}-2z^{-1}}$$

$\rightarrow$ to have z with negative powers

z^{-1} (max positive power in denominator and Numerator) as we have $(x[n-a] \leftrightarrow z^{-a}X(z))$

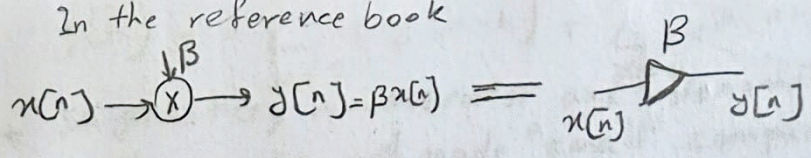
Software - Based Implementation (slides)

Hardware - Based $\rightarrow$ On FPGA $\rightarrow$ We need to know the block diagram.

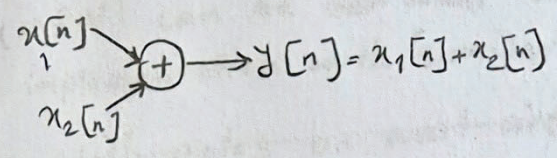
In block diagram we implement a system by using 3 blocks

In the reference book

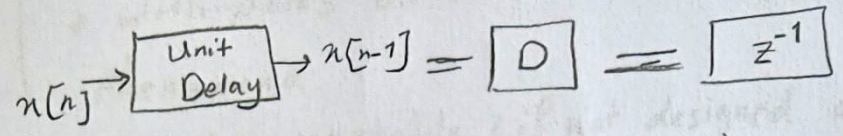
Multiplier



Adder



Unit-Delay



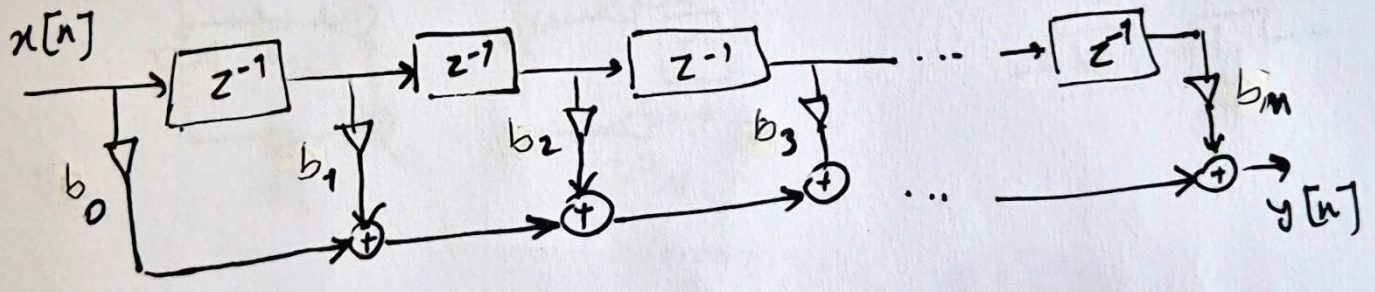
FIR Filters

$$y[n] = \sum_{k=0}^M (b_k) x[n-k] = \sum_{k=0}^M h[k] x[n-k]$$

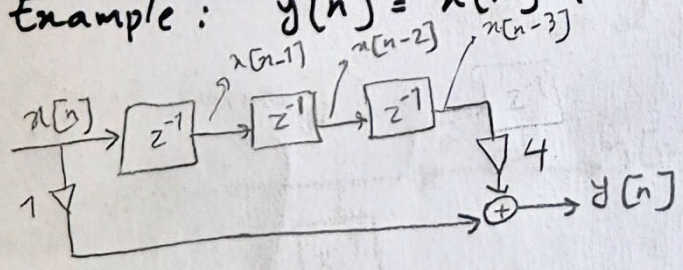
book's notation b_k

$\downarrow$ $k=0$
 $x[n] * h[n]$

- Properties:
- + Always stable (m is limited) or Poles at $z=0$
 - + Can be made with linear phase (having symmetric Property for Impulse Response)
 - In practice, the impulse response (or the filter order) needs to be long to fulfill requirements.



- Example: $y[n] = x[n] + 4x[n-3]$

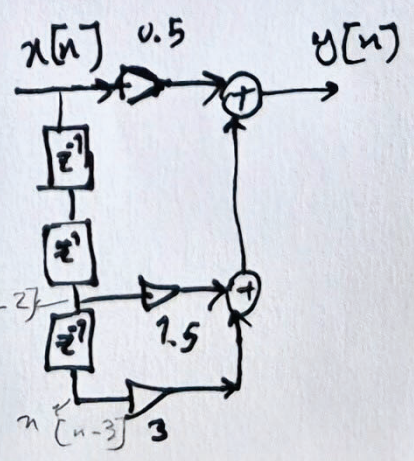


- Example: The system function?

$$y[n] = 0.5x[n] + 1.5x[n-2] + 3x[n-3]$$

$Z_{-Trans} \rightarrow Y(z) = 0.5X(z) + 1.5X(z) \cdot z^{-2} + 3z^{-3}X(z)$

$$\rightarrow H(z) = \frac{Y(z)}{X(z)} = 0.5 + 1.5z^{-2} + 3z^{-3}$$



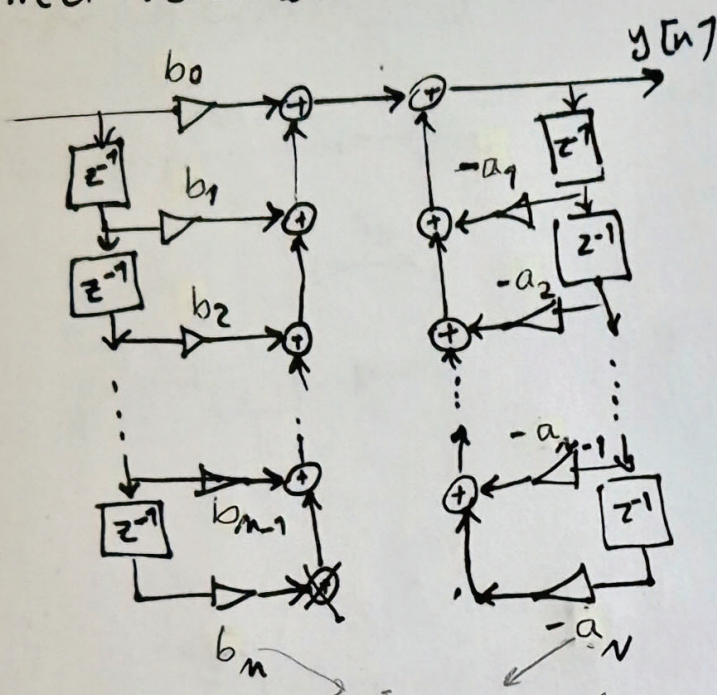
IIR Filters:

$$y[n] = - \sum_{k=1}^N a_k y[n-k] + \sum_{k=0}^M b_k x[n-k]$$

Properties →

- + In practice, numerator and denominator orders (M & N) can be made "small" compared to FIR implementation with "similar" properties.
- + With poles on unit circle, we can include resonance phenomena
- can be unstable (if not designed correctly)
- cannot have a linear phase.

Direct Form I:



In your reference book a_k are used as coefficients of $y[n-k]$

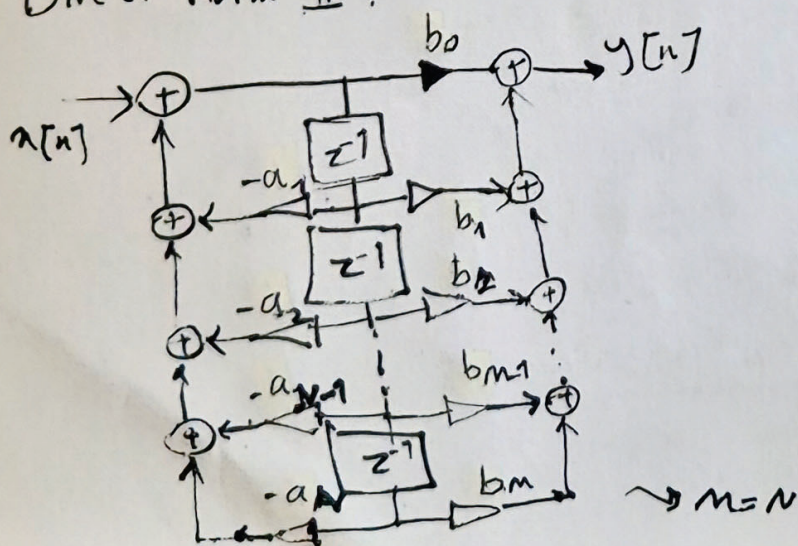
Commutative
↑ Property

Conv.
Property

$$h_t[n] = h_1[n] * h_2[n] = h_2[n] * h_1[n]$$

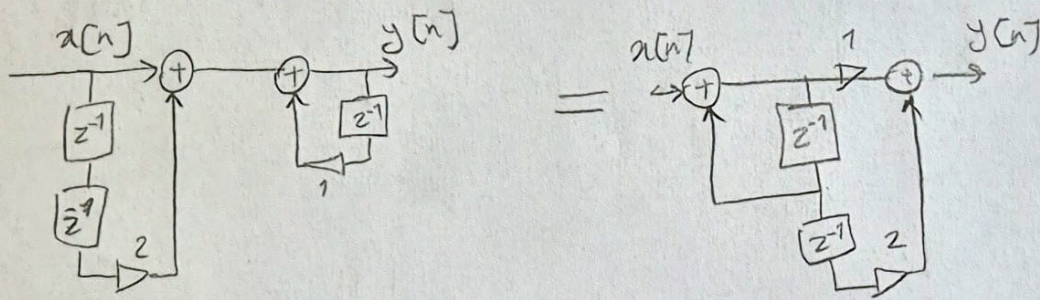
[TI]
→ Property

Direct Form II:

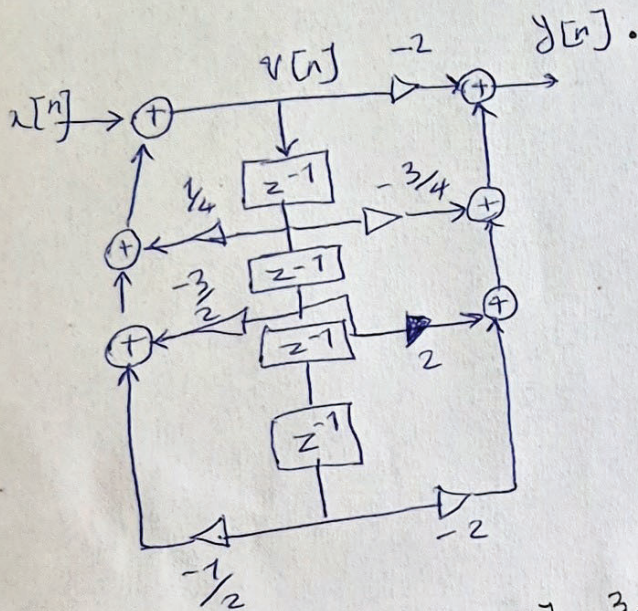


Example $y[n] = x[n] + 2x[n-2] + y[n-1] \rightarrow \text{Form II? } \textcircled{4}$

First Form I:



Example: A system is given in the following format:



Determine the system function, $H(z)$, by using $v[n]$:

$$y[n] = -2v[n] + \frac{3}{4}v[n-1] + 2v[n-2] - 2v[n-4] \quad \textcircled{2}$$

$$v[n] = x[n] + \frac{1}{4}v[n-1] - \frac{3}{2}v[n-2] - \frac{1}{2}v[n-4] \quad \textcircled{1}$$

Let's use $\textcircled{1}$
Z-Trans.

$$V(z) = \frac{1}{1 - \frac{1}{4}z^{-1} + \frac{3}{2}z^{-2} + \frac{1}{2}z^{-4}} X(z) \quad \textcircled{3}$$

$$\begin{aligned} \textcircled{2} \text{ Z-Trans } Y(z) &= -2V(z) - \frac{3}{4}z^{-1}V(z) + 2z^{-2}V(z) - 2V(z)z^{-4} \\ Y(z) &= V(z) \cdot (-2 - \frac{3}{4}z^{-1} + 2z^{-2} - 2z^{-4}) \\ &= \frac{(-2 - \frac{3}{4}z^{-1} + 2z^{-2} - 2z^{-4})}{1 - \frac{1}{4}z^{-1} + \frac{3}{2}z^{-2} + \frac{1}{2}z^{-4}} X(z) \\ H(z) &= \frac{Y(z)}{X(z)} = \frac{-2 - \frac{3}{4}z^{-1} + 2z^{-2} - 2z^{-4}}{1 - \frac{1}{4}z^{-1} + \frac{3}{2}z^{-2} + \frac{1}{2}z^{-4}} \end{aligned}$$